FinTrust Digital Bank — Data Analytics Track

Week 4, Part A: Final Data & Analysis Review

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This document reviews all Data Analytics track work completed in Weeks 1–3 against the six Part A checkpoints, confirms what has already been verified, and records the corrections and improvements made in preparation for the final Week 4 deliverables.

1. Data Quality Issues

Status: Confirmed, re-verified for Week 4. The Week 2 Data Quality & Cleaning Workbook originally identified 96 missing Device_Type values and 96 missing Location values (0.8% each) in the Transaction_Data file, with zero duplicate Customer_ID or Transaction_ID values and no negative amounts or implausible ages. These checks were re-run against the same source files for Week 4 and produced identical results — no data drift, since the underlying CSVs are static.

Action taken: Missing Device_Type and Location values remain imputed as "Unknown" in the cleaned dataset used for all downstream SQL, Python, and Power BI work. No further cleaning action was required.

Data import issue found in Week 4 (SQL Server only). Reconciling totals showed that SUM(Amount_NGN) in SQL Server was 554,157,109.34, against 560,477,354.85 in the source CSV. The CSV has no missing amounts. Exactly 128 transactions whose amounts are whole numbers in the CSV (no decimal point) were imported into SQL Server as NULL, and their total (NGN 6,320,246) equals the gap. The column had also been stored as float. A repair script (FinTrust_SQL_Amount_Repair.sql) restores the 128 amounts and converts the column to DECIMAL(18,2). Row counts, rates and percentages were never affected; only SQL totals and averages of the amount were about 1.1% low.

2. Calculation Accuracy

Status: Confirmed for Python and Power BI; SQL value totals corrected. Headline figures - total transaction value (NGN 560.48M), average transaction value (NGN 46,706.45 / NGN 46.71K) and success rate (90.47%) - were recomputed from the raw CSVs in Python and match the Power BI cards. Counts and rates (success rate, risk rate, channel and segment counts) also matched in SQL. SQL value totals did not match until the import issue above was repaired, so reconciling each tool against the source data was what exposed the problem.

Correction documented: Record counts are reported as 1,501 (Customer_Data) and 12,001 (Transaction_Data), i.e. including the header row. This is a stated convention, not a counting error — it is now explicitly documented here so a reviewer does not mistake it for a miscount. The effective record counts used in all analysis remain 1,500 customers and 12,000 transactions.

3. SQL Query Validation

Status: Confirmed after repair. The 8 Week 2 business questions were run in SQL Server, so any result that sums or averages the amount was about 1.1% low before the repair (for example Everyday segment value NGN 258.7M instead of NGN 261.5M). Counts and rates were unaffected. The 8 Week 3 advanced queries were

checked against the raw CSV files using DuckDB. The final script's results are the raw-CSV figures, and after the repair SQL Server reproduces them (total NGN 560,477,355.91, within NGN 1.06 of the CSV total because of rounding in the original float import). The Week 2 and Week 3 files are left as originally submitted, and this document records the cause.

Known data-type issue, resolved and documented: SQL Server auto-typed Risk_Review_Flag and International_Transaction as bit (boolean) on import, rather than text. All SQL queries compare these fields to 1/0 rather than 'Yes'/'No'. This is noted here so the same issue does not recur when queries are reviewed or extended in Week 4.

4. Python Analysis Reproducibility

Status: Reproducible on the original machine; limitation documented for Week 4. The Jupyter notebook (Fintrust_Week2_Data_Analysis.ipynb) connects live to SQL Server via pyodbc rather than reading from a static exported file. All 10 visualizations (5 from Week 2, 5 from Week 3) were verified by matching their output numbers against the equivalent SQL results.

Effect of the import issue on the notebook: the notebook loads its data from SQL Server, so it also received the 128 NULL amounts. Charts built on the amount (histogram, spend vs frequency, success rate by value range) shift very slightly after the repair. The CSV snapshots are created after the repair so they contain the correct values.

Limitation to document in Part F: because the notebook depends on a live pyodbc connection to a local SQL Server instance, it is reproducible by the author but not portable without the CSV snapshots. This is stated plainly as a documented limitation rather than left implicit.

Improvement made for Week 4: the notebook's data-loading cell saves a CSV snapshot of both tables from SQL Server and falls back to those CSVs when no database is available. A final "Restart & Run All" is performed before submission to confirm no hidden variable state is propping up any chart.

5. KPI Definitions

Status: Previously implemented as dashboard cards; not previously documented as a standalone table. The table below is the first standalone KPI documentation produced for this project, consolidating the KPIs already live on the Power BI dashboard.

KPI	Definition / Formula	Business Purpose
Total Customers	Distinct count of Customer_ID	Measures the size of the active customer base
Total Transactions	Count of Transaction_ID	Measures overall platform activity volume
Total Transaction Value	SUM(Amount_NGN)	Measures the total value flowing through FinTrust
Average Transaction Value	SUM(Amount_NGN) / COUNT(Transaction_ID)	Indicates typical transaction size; flags right-skew

KPI	Definition / Formula	Business Purpose
Success Rate %	DIVIDE(COUNTROWS(FILTER(Status="Successful")), COUNTROWS(all transactions))	Tracks transaction reliability; corrected from an earlier AVERAGE()-on-text error to a proper COUNTROWS/FILTER/DIVIDE pattern
Average Daily Value	SUM(Amount_NGN) / DISTINCTCOUNT(transaction date)	Added in Week 4 so monthly trends are compared fairly (February has 28 days, so raw monthly totals mislead)
Risk Review Rate %	DIVIDE(COUNTROWS(FILTER(Risk_Review_Flag="Yes")), COUNTROWS(all transactions))	Tracks the share of transactions requiring manual risk review

Correction documented: the Risk Review Rate % measure originally used AVERAGE() on a text field ("Yes"/"No"), which does not calculate a rate. It was corrected in Week 3 to the COUNTROWS/FILTER/DIVIDE pattern shown above, the DAX-correct way to calculate a rate over a text-categorical field. Success Rate % was built with the same pattern from the start.

6. Findings Supported by Evidence

Status: Confirmed for the numbers; three interpretations revised. Week 3's five validated findings followed a Finding, Evidence, Validation, Business Meaning structure and were cross-checked across SQL, Python and Power BI. In Week 4, significance tests (Part E) showed that three conclusions went further than the evidence supports: (1) ATM + International (46.0%) rests on only 63 transactions (95% interval about 34% to 58%), so international status as a whole is the supported driver (36.9% vs 18.9%); (2) failures and reversals are not concentrated in USSD, they run at about 8% across all channels (chi-square p = 0.07); (3) the February dip is a calendar effect, because February has 28 days and transactions per day are identical across the three months. The numbers behind these findings were correct. Only the conclusions drawn from them changed.

7. Summary of Corrections & Improvements for Week 4

The table below consolidates every correction or improvement identified during this review, as required by Part A (items 6 to 10 were identified during Week 4 validation and reconciliation).

#	Area	Issue Identified	Action Taken / Planned
1	Documentation	Record counts (1,501 / 12,001) not explained as a header-inclusive convention	Explicitly documented in this review and in final Week 4 documentation
2	Python notebook	Live pyodbc/SQL Server dependency limits portability for a grader	Data-loading cell now saves CSV snapshots and falls back to them without SQL Server; limitation documented; Restart & Run All before final submission
3	KPI documentation	KPIs existed only as dashboard cards, with no standalone definition table	Standalone KPI table produced (Section 5 above) for inclusion in final Week 4 documentation
4	Power BI DAX	Risk Review Rate % originally used AVERAGE() on a text field	Corrected in Week 3 to COUNTROWS/FILTER/DIVIDE pattern; re-confirmed correct for Week 4
5	Data quality	Missing Device_Type / Location values in Transaction_Data	Re-verified as still correctly imputed as "Unknown"; no drift since Week 2
6	Analysis conclusion	ATM + International (46.0%) presented as the highest-risk hotspot	Based on 63 transactions; revised to: international status overall drives risk (36.9% vs 18.9%)
7	Analysis conclusion	Failures and reversals described as concentrated in USSD	Not statistically supported (p = 0.07); revised to a platform-wide rate of about 8%
8	Analysis conclusion	February dip treated as a possible genuine fluctuation	Calendar effect (28 days); daily value is flat to slightly rising; Average Daily Value measure added
9	Power BI dashboard	Key Highlights repeated revised conclusions; trend chart axis exaggerated the dip; pie chart labels overlapped	Highlights rewritten; trend changed to a zero-based column chart; status pie replaced with a percentage bar chart; risk-by-channel changed to 100% stacked; labels, axis titles and tab names standardised
10	SQL data import	128 whole-number amounts imported as NULL; SQL total 554.16M vs 560.48M in the CSV; column stored as float	Found by reconciling SUM(Amount_NGN) to the source; repaired with FinTrust_SQL_Amount_Repair.sql (restore 128 amounts, DECIMAL(18,2)); notebook snapshots rebuilt after the repair

Conclusion: the review found one data-handling error and three overstated conclusions. The error was a SQL Server import that turned 128 whole-number amounts into NULL, which made SQL totals about 1.1% low; it was found by reconciling against the source file and repaired. The three conclusions (items 6 to 8) were revised after

significance testing and are documented in Parts D and E. Calculations in Python and Power BI were correct, and counts and rates were consistent across all three tools.